



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0373



Agricultural Machinery – Power Sprayer for Mango BLUMAX BM-30A

Test Report Number	2018 – 0373
Valid Until	July 6, 2023
Agricultural Machinery	Power Sprayer for Mango
Type	Plunger, Three Cylinder, Frame-mounted
Brand	BLUMAX
Model	BM-30A
Test Requested by	BRIXTON CONSTRUCTION & INDUSTRIAL SUPPLY CORP. Room 202, Armon's Bldg., Kamias Road Cor. Anonas St., East Kamias, Quezon City

SUMMARY

Manufacturing and Performance Requirements	Based on PAES157:2011	AMTEC Test
Pump	Positive Displacement	Positive Displacement
Spray Hose	Polyvinylchloride or better; Inside Diameter: at least 10 mm Length: 15 m – 200 m	Polyvinylchloride; Inside Diameter: 5 mm Length: 51.36 m
Intake Hose	Polyvinylchloride or better; Inside Diameter: at least 16 mm Provided with filter	Polyvinylchloride; Inside Diameter: 13 mm; Length: 2.33
Lance	Non-corrosive steel or better; Length: at least 1 m	Non-corrosive steel; Length: 0.64 m
Nozzle	Non-corrosive; Detachable; & Adjustable	Non-corrosive, removable, and non-adjustable
Pressure Gauge	Shall have	Present
Strainer/Filter, holes/cm ²	100	204
Discharge Rate, Lpm	At least 15	13.7
Operating Pressure	1.21 MPa – 1.75 MPa (175 psi – 250 psi)	2.97 MPa to 3.45 MPa
Noise Level, dB(A)	92* maximum	93.1

Note: *Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines. 2013.

OBSERVATIONS

1. Operator's Manual (based on PAES 102:2000)	Not Provided
2. Previous Valid Test Reports of Similar Brand and Model	None
3. Prime mover Used	4.85 kW BLUMAX BMG 168FB Gasoline Engine
4. Manufacturer	WEIMA POWER MACHINE CP., LTD. Chongqing, China
5. Serial Number	None
6. Number of Operator	One person
7. Other Remarks	No breakdown occurred during the test. Safety cover was not provided for the belt and pulley transmission system.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date/s of Test	May 18, 2018
Location of Test	AMTEC Testing Laboratory, UP Los Baños, College, Laguna
Items Tested	Maximum Pressure, Discharge Rate, Sprayer Range, Fuel Consumption, Speed of Components, and Noise Level

METHODS OF TEST

The power sprayer was tested based on the Philippine Agricultural Engineering Standards, PAES 158:2011 Agricultural Machinery – Power Sprayer for Mango - Methods of Test.

DESCRIPTION OF THE MACHINE

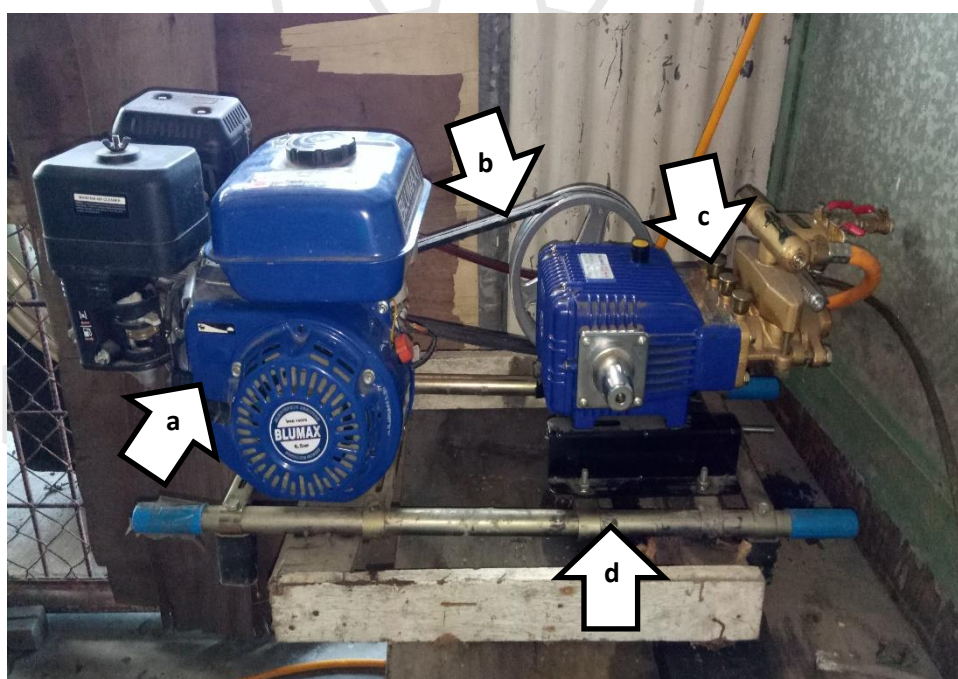


Figure 1. Parts of the BLUMAX BM-30A Powered by 4.85kW BLUMAX BMG 168FB Gasoline Engine: (a) Prime mover; (b) Belt and pulley transmission; (c) Plungers; and (d) Main frame.

MACHINE SPECIFICATIONS

ITEM	Manufacturer's Specification	Verification by AMTEC
1 overall dimensions		
1.1 height, mm	ND	860
1.2 length, mm	ND	340
1.3 width, mm	ND	605
2 spray hose		
2.1 material	ND	Polyvinylchloride
2.2 inside diameter, mm	ND	5
2.3 length, mm	ND	51360
3 suction hose		
3.1 material	ND	Polyvinylchloride
3.2 inside diameter, mm	ND	13
3.3 length, mm	ND	2330
4 lance		
4.1 material	ND	Non-corrosive steel
4.2 thickness, mm	ND	NM
4.3 length, mm	ND	635
5 nozzle		
5.1 diameter, mm	ND	NM
5.2 material	ND	Non-corrosive steel
6 prime mover		
6.1 type	ND	Four – stroke, Gasoline Engine
6.2 power rating, kW	ND	4.85
6.3 type of starting	ND	Recoil starter
6.4 revolutions per minute	ND	1426 – 3610*
7 mounting		
		Tubular steel frame
8 width of spray, m		
		NM
9 sprayer range, m		
9.1 horizontal, m	ND	19 – 22*
9.2 vertical, m	ND	10 – 12*
10 angle of spray, deg		
		NM
11 pump		
11.1 operating pressure, MPa	ND	2.97 – 3.45*
11.2 type	ND	Positive Displacement

Note: ND - No Data; NM – Not Measured, *Based on actual test data.

RESULTS

Test Condition

Ambient Temperature (°C) : 32.3
 Atmospheric Pressure (mb) : 1013
 Relative Humidity (%) : 64.4

Table 2. Performance Test Results of BLUMAX BM-30A Power Sprayer Powered by 4.85 kW BLUMAX BMG 168FB Gasoline Engine

Shaft Speed (rpm)		Fuel Consumption (L/h)	Pressure (KPa)	Discharge (L/min)	Spray Range (m)		Noise Level [dB(A)]	Output Power (kW)	Input Power (kW)	System Efficiency (%)
Engine Shaft	Sprayer Pump Shaft				Horizontal	Vertical				
3610	1623	1.35	3450.0	13.57	22.0	12.00	93.10	0.780	12.52	6.23
3426	1555	1.20	3450.0	13.70	22.0	12.00	92.10	0.788	11.12	7.08
3210	1386	1.12	3450.0	13.65	22.0	12.00	91.30	0.785	10.38	7.57
3015	1248	1.08	3450.0	13.59	22.0	12.00	89.60	0.782	9.96	7.85
2821	1138	0.95	3450.0	13.47	22.0	12.00	89.50	0.775	8.77	8.83
2602	1178	0.88	3450.0	13.51	22.0	12.00	88.20	0.777	8.14	9.55
2411	1085	0.77	3312.0	13.29	21.0	11.00	86.40	0.734	7.14	10.28
2216	1005	0.72	3312.0	13.22	21.0	11.00	85.60	0.730	6.66	10.96
2024	900	0.65	3105.0	13.14	20.0	10.00	85.70	0.680	6.03	11.28
1829	824	0.61	3105.0	12.97	20.0	10.00	85.10	0.672	5.60	11.99
1624	692	0.54	2967.0	12.72	19.0	10.00	83.40	0.629	5.00	12.59
1426	473	0.50	2967.0	13.50	19.0	10.00	82.40	0.668	4.67	14.29

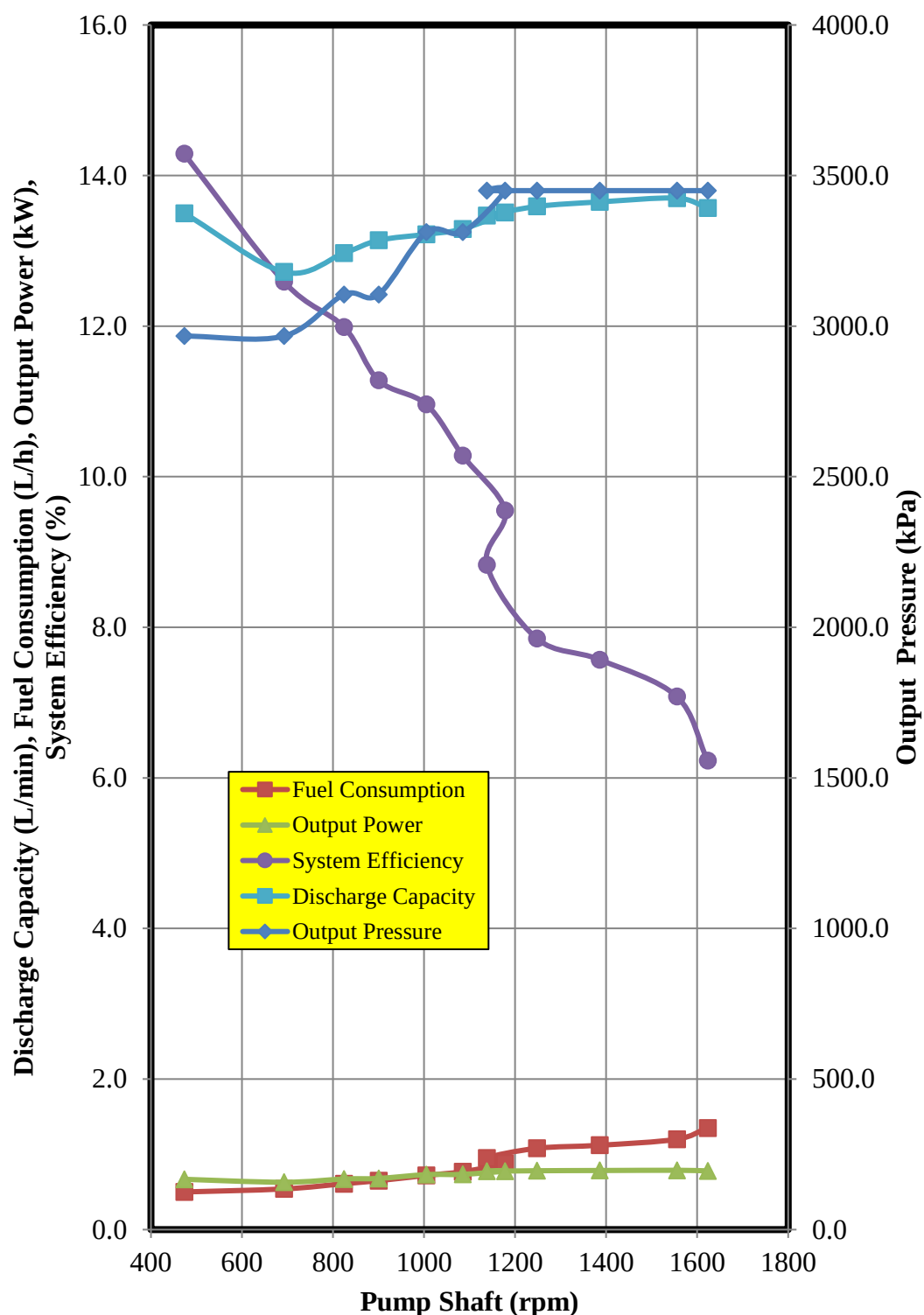


Figure 2. The Performance Characteristics Curves of the BLUMAX BM-30A Power Sprayer Powered by 4.85 kW BLUMAX BMG 168FB Gasoline Engine

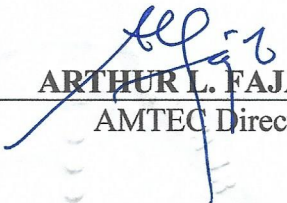
Prepared by:


FATIMA JOY J. RAYTANA
Test Engineer

Verified by:


ROMULO E. EUSEBIO
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

SEP 03 2019
Date

REMINDER

This test report should not be used for advertisement or for promotion of the product but only for evaluation and verification purposes and should not be reproduced in whole or in part without the permission of AMTEC.

For inquiries regarding the test report, you can visit us at our office, contact us at +639 772 047 236 or +639 214 007 137, or send us a message at amtec.uplb@up.edu.ph or visit our official website at amtec.uplb.edu.ph.